

Imię i nazwisko: Paweł Borowiecki

Nazwa programu: GNU Hello

GNU Hello to program, który wyświetla tekst powitalny. Tekst może wyświetlać się w różnych językach. GNU Hello zostało stworzone w języku C.

Przebieg procesu instalacji oprogramowania, zależności i wynik działania:

Na początku należy przejść do katalogu Pobrane i tam pobrać oprogramowanie ze strony producenta, za pomocą komendy wget.

```
pawel@myubuntu:~/Pobrane$ wget https://ftp.gnu.org/gnu/hello/hello-2.12.3.tar.gz
--2026-05-23 21:55:47-- https://ftp.gnu.org/gnu/hello/hello-2.12.3.tar.gz
Translacja ftp.gnu.org (ftp.gnu.org)... 209.51.188.20, 2001:470:142:3::b
Łączenie się z ftp.gnu.org (ftp.gnu.org)[209.51.188.20]:443... połączono.
Żądanie HTTP wysłano, oczekiwanie na odpowiedź... 200 OK
Długość: 1189208 (1,1M) [application/x-gzip]
Zapis do: 'hello-2.12.3.tar.gz'

hello-2.12.3.tar.gz      100%[=====>]  1,13M  1,61MB/s   w 0,7s

2026-05-23 21:55:48 (1,61 MB/s) - zapisano 'hello-2.12.3.tar.gz' [1189208/1189208]

pawel@myubuntu:~/Pobrane$
```

Potem można sprawdzić, za pomocą polecenia ls, czy pakiet został pobrany i należy rozpakować go, za pomocą komendy tar z opcjami xzvf.

```
pawel@myubuntu:~/Pobrane$ ls
anything.txt  aria2-1.37.0.tar.gz  nothing.txt  sth.txt
aria2-1.37.0  hello-2.12.3.tar.gz  Somewhere
pawel@myubuntu:~/Pobrane$ tar -xzf hello-2.12.3.tar.gz
```

```
hello-2.12.3/po/stamp-po
hello-2.12.3/po/sv.gmo
hello-2.12.3/po/sv.po
hello-2.12.3/po/ta.gmo
hello-2.12.3/po/ta.po
hello-2.12.3/po/th.gmo
hello-2.12.3/po/th.po
hello-2.12.3/po/tr.gmo
hello-2.12.3/po/tr.po
hello-2.12.3/po/uk.gmo
hello-2.12.3/po/uk.po
hello-2.12.3/po/vi.gmo
hello-2.12.3/po/vi.po
hello-2.12.3/po/zh_CN.gmo
hello-2.12.3/po/zh_CN.po
hello-2.12.3/po/zh_TW.gmo
hello-2.12.3/po/zh_TW.po
hello-2.12.3/src/
hello-2.12.3/src/hello.c
hello-2.12.3/src/system.h
hello-2.12.3/tests/
hello-2.12.3/tests/atexit-1
hello-2.12.3/tests/greeting-1
hello-2.12.3/tests/greeting-2
hello-2.12.3/tests/hello-1
hello-2.12.3/tests/last-1
hello-2.12.3/tests/operand-1
hello-2.12.3/tests/traditional-1
nawel@myubuntu:~/Pobrane$
```

Następnie należy przejść do katalogu źródłowego pakietu, sprawdzić czy są zainstalowane zależności, jakich wymaga pakiet (polecenie `dpkg -s`) i je ewentualnie doinstalować.

W moim przypadku pakiety, które sprawdziłem i ewentualnie doinstalowałem to: automake, autoconf, gnuilib, gettext oraz build-essential.

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ dpkg -s automake
Package: automake
Status: install ok installed
Priority: optional
Section: devel
Installed-Size: 1581
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: all
Multi-Arch: foreign
Source: automake-1.16
Version: 1:1.16.5-1.3
Provides: automake-1.16, automaken
Depends: autoconf, autotools-dev
Suggests: autoconf-doc, gnu-standards
Description: Tool for generating GNU Standards-compliant Makefiles
 Automake is a tool for automatically generating `Makefile.in's from
 files called `Makefile.am'.
.
The goal of Automake is to remove the burden of Makefile maintenance
from the back of the individual GNU maintainer (and put it on the back
of the Automake maintainer).
.
The `Makefile.am' is basically a series of `make' macro definitions
(with rules being thrown in occasionally). The generated
`Makefile.in's are compliant with the GNU Makefile standards.
.
Automake 1.16 fails to work in a number of situations that Automake
1.11, and 1.15 did, so some previous versions are available as separate
packages.
Original-Maintainer: Eric Dorland <eric@debian.org>
Homepage: https://www.gnu.org/software/automake/
```

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ dpkg -s autoconf
Package: autoconf
Status: install ok installed
Priority: optional
Section: devel
Installed-Size: 2032
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: all
Multi-Arch: foreign
Version: 2.71-2
Depends: perl (>= 5.005), m4 (>= 1.4.13), debianutils (>= 1.8)
Recommends: automake | automaken
Suggests: autoconf-archive, gnu-standards, autoconf-doc, libtool, gettext
Breaks: gettext (< 0.10.39), pkg-config (< 0.25-1.1)
Conffiles:
  /usr/share/doc/autoconf/emacs/site-start.d/50autoconf.el 297521889d690871ec9d89c5eeff745a
Description: automatic configure script builder
 The standard for FSF source packages. This is only useful if you
 write your own programs or if you extensively modify other people's
 programs.
 .
 For an extensive library of additional Autoconf macros, install the
 `autoconf-archive' package.
 .
 This version of autoconf is not compatible with scripts meant for
 Autoconf 2.13 or earlier.
Original-Maintainer: Debian QA Group <packages@qa.debian.org>
Homepage: https://www.gnu.org/software/autoconf/
```

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ dpkg -s gnuilib
dpkg-query: pakiet "gnuilib" nie jest zainstalowany i informacje o nim nie są dostępne
Użyj dpkg --info (= dpkg-deb --info), aby zobaczyć informacje o pakiecie.
```

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ sudo apt install gnuilib
[sudo] hasło użytkownika pawel:
Niestety, proszę spróbować ponownie.
[sudo] hasło użytkownika pawel:
Czytanie list pakietów... Gotowe
Budowanie drzewa zależności... Gotowe
Odczyt informacji o stanie... Gotowe
Następujące pakiety zostały zainstalowane automatycznie i nie są już więcej wymagane:
  bridge-utils ubuntu-fan
Aby je usunąć należy użyć "sudo apt autoremove".
Zostaną zainstalowane następujące dodatkowe pakiety:
  bison gperf libtext-unidecode-perl tex-common texinfo
Sugerowane pakiety:
  bison-doc clisp texlive-base texlive-latex-base texlive-plain-generic
  texlive-fonts-recommended
Zostaną zainstalowane następujące NOWE pakiety:
  bison gnuilib gperf libtext-unidecode-perl tex-common texinfo
0 aktualizowanych, 6 nowo instalowanych, 0 usuwanych i 23 nieaktualizowanych.
Konieczne pobranie 7 192 kB archiwów.
Po tej operacji zostanie dodatkowo użyte 62,2 MB miejsca na dysku.
Kontynuować? [T/n] T
```

```

(Odczytywanie bazy danych ... 212130 plików i katalogów obecnie zainstalowanych.)
Przygotowywanie do rozpakowania pakietu .../0-tex-common_6.17_all.deb ...
Rozpakowywanie pakietu tex-common (6.17) ...
Wybieranie wcześniej niewybranego pakietu bison.
Przygotowywanie do rozpakowania pakietu .../1-bison_2%3a3.8.2+dfsg-1build1_amd64.deb ...
Rozpakowywanie pakietu bison (2:3.8.2+dfsg-1build1) ...
Wybieranie wcześniej niewybranego pakietu gnuilib.
Przygotowywanie do rozpakowania pakietu .../2-gnuilib_20210822~d383792-1_all.deb ...
Rozpakowywanie pakietu gnuilib (20210822~d383792-1) ...
Wybieranie wcześniej niewybranego pakietu gperf.
Przygotowywanie do rozpakowania pakietu .../3-gperf_3.1-1build1_amd64.deb ...
Rozpakowywanie pakietu gperf (3.1-1build1) ...
Wybieranie wcześniej niewybranego pakietu libtext-unidecode-perl.
Przygotowywanie do rozpakowania pakietu .../4-libtext-unidecode-perl_1.30-1_all.deb ...
Rozpakowywanie pakietu libtext-unidecode-perl (1.30-1) ...
Wybieranie wcześniej niewybranego pakietu texinfo.
Przygotowywanie do rozpakowania pakietu .../5-texinfo_6.8-4build1_amd64.deb ...
Rozpakowywanie pakietu texinfo (6.8-4build1) ...
Konfigurowanie pakietu tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Konfigurowanie pakietu gnuilib (20210822~d383792-1) ...
Konfigurowanie pakietu bison (2:3.8.2+dfsg-1build1) ...
update-alternatives: użycie /usr/bin/bison.yacc jako dostarczającego /usr/bin/yacc (yacc) w trybie automatycznym
Konfigurowanie pakietu libtext-unidecode-perl (1.30-1) ...
Konfigurowanie pakietu texinfo (6.8-4build1) ...
Przetwarzanie wyzwalaczy pakietu man-db (2.10.2-1)...
Przetwarzanie wyzwalaczy pakietu install-info (6.8-4build1)...
pawel@myubuntu: ~/Pobrane/hello-2.12.3$

```

```

pawel@myubuntu: ~/Pobrane/hello-2.12.3$ dpkg -s gettext
Package: gettext
Status: install ok installed
Priority: optional
Section: devel
Installed-Size: 2636
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Multi-Arch: foreign
Version: 0.21-4ubuntu4
Depends: libc6 (>= 2.34), libgomp1 (>= 6), libunistring2 (>= 1.0), libxml2 (>= 2.9.1), gettext-base, dpkg (>= 1.15.4) | install-info
Recommends: curl | wget | lynx
Suggests: gettext-doc, autopoint, libasprintf-dev, libgettextpo-dev
Breaks: autopoint (<= 0.17-11)
Description: GNU Internationalization utilities
 Interesting for authors or maintainers of other packages or programs
  which they want to see internationalized.
Homepage: https://www.gnu.org/software/gettext/
Original-Maintainer: Santiago Vila <sanvila@debian.org>

```

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ dpkg -s build-essential
Package: build-essential
Status: install ok installed
Priority: optional
Section: devel
Installed-Size: 21
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 12.9ubuntu3
Depends: libc6-dev | libc-dev, gcc (>= 4:10.2), g++ (>= 4:10.2), make, dpkg-dev (>= 1.17.11)
Description: Informational list of build-essential packages
 If you do not plan to build Debian packages, you don't need this
 package. Starting with dpkg (>= 1.14.18) this package is required
 for building Debian packages.
.
This package contains an informational list of packages which are
considered essential for building Debian packages. This package also
depends on the packages on that list, to make it easy to have the
build-essential packages installed.
.
If you have this package installed, you only need to install whatever
a package specifies as its build-time dependencies to build the
package. Conversely, if you are determining what your package needs
to build-depend on, you can always leave out the packages this
package depends on.
.
This package is NOT the definition of what packages are
build-essential; the real definition is in the Debian Policy Manual.
This package contains merely an informational list, which is all
```

Kolejnym krokiem jest uruchomienie skryptu konfiguracji.


```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ ./configure
checking for a BSD-compatible install... /usr/bin/install -c
checking whether sleep supports fractional seconds... yes
checking filesystem timestamp resolution... 0.01
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... no
checking for mawk... mawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking xargs -n works... yes
checking whether UID '1000' is supported by ustar format... yes
checking whether GID '1000' is supported by ustar format... yes
checking how to create a ustar tar archive... gnutar
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking whether gcc understands -c and -o together... yes
checking whether the compiler is clang... no
checking for compiler option needed when checking for declarations... none
checking whether make supports the include directive... yes (GNU style)
checking dependency style of gcc... gcc3
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
```

```
checking for struct stat.st_atim.tv_nsec... yes
checking whether struct stat.st_atim is of type struct timespec... yes
checking for struct stat.st_birthtimespec.tv_nsec... no
checking for struct stat.st_birthtimensec... no
checking for struct stat.st_birthtim.tv_nsec... no
checking for va_copy... yes
checking for vasprintf... yes
checking for working strerror function... yes
checking for working strlen... yes
checking for variable-length arrays... yes
checking whether wwidth is declared... yes
checking whether wwidth works reasonably in UTF-8 locales... yes
checking whether NLS is requested... yes
checking for msgfmt... /usr/bin/msgfmt
checking for gmsgfmt... /usr/bin/msgfmt
checking for xgettext... /usr/bin/xgettext
checking for msgmerge... /usr/bin/msgmerge
checking for CFPReferencesCopyAppValue... no
checking for CFLocaleCopyCurrent... no
checking for GNU gettext in libc... yes
checking whether to use NLS... yes
checking where the gettext function comes from... libc
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating po/Makefile.in
config.status: creating config.h
config.status: executing depfiles commands
config.status: executing po-directories commands
config.status: creating po/POTFILES
config.status: creating po/Makefile
nawel@myubuntu:~/Pobrane/hello-2.12.3$
```

Następnie należy dokonać kompilacji pakietu, za pomocą komendy make.


```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ make
/usr/bin/mkdir -p 'lib'
sed -e 1h -e '1s,.*,/* DO NOT EDIT! GENERATED AUTOMATICALLY! */,' -e 1G \
-e 's|@'|HAVE_ALLOCA_H'|@|1|g' \
./lib/alloca.in.h > lib/alloca.h-t
mv lib/alloca.h-t lib/alloca.h
/usr/bin/mkdir -p 'lib'
{ echo '/* DO NOT EDIT! GENERATED AUTOMATICALLY! */'; \
echo '#if HAVE_WINSOCK2_H'; \
echo '# include <winsock2.h> /* avoid mingw pollution on DATADIR */'; \
echo '#endif'; \
printf '#define PREFIX %s\n' \" /usr/local\"; \
printf '#define EXEC_PREFIX %s\n' \" /usr/local\"; \
printf '#define BINDIR %s\n' \" /usr/local/bin\"; \
printf '#define SBINDIR %s\n' \" /usr/local/sbin\"; \
printf '#define LIBEXECDIR %s\n' \" /usr/local/libexec\"; \
printf '#define DATAROOTDIR %s\n' \" /usr/local/share\"; \
printf '#define DATADIR %s\n' \" /usr/local/share\"; \
printf '#define SYSCONFDIR %s\n' \" /usr/local/etc\"; \
printf '#define SHAREDSTATEDIR %s\n' \" /usr/local/com\"; \
printf '#define LOCALSTATEDIR %s\n' \" /usr/local/var\"; \
printf '#define RUNSTATEDIR %s\n' \" /usr/local/var/run\"; \
printf '#define INCLUDEDIR %s\n' \" /usr/local/include\"; \
printf '#define OLDINCLUDEDIR %s\n' \" /usr/include\"; \
printf '#define DOCDIR %s\n' \" /usr/local/share/doc/hello\"; \
printf '#define INFODIR %s\n' \" /usr/local/share/info\"; \
printf '#define HTMLDIR %s\n' \" /usr/local/share/doc/hello\"; \
printf '#define DVIDIR %s\n' \" /usr/local/share/doc/hello\"; \
printf '#define PDFDIR %s\n' \" /usr/local/share/doc/hello\"; \
printf '#define PSDIR %s\n' \" /usr/local/share/doc/hello\"; \
printf '#define LIBDIR %s\n' \" /usr/local/lib\"; \
printf '#define LISPDIR %s\n' \" /usr/local/share/emacs/site-lisp\"; \
```

```
Po
gcc -DLOCALEDIR=\"/usr/local/share/locale\" -DHAVE_CONFIG_H -I. -Ilib -I./lib -Isrc -I./src -
Wno-cast-qual -Wno-conversion -Wno-float-equal -Wno-sign-compare -Wno-undef -Wno-unused-function
-Wno-unused-parameter -Wno-float-conversion -Wimplicit-fallthrough -Wno-pedantic -Wno-sign-conv
ersion -Wno-type-limits -Wno-unused-const-variable -Wno-unsuffixed-float-constants -Wno-error -g
-O2 -MT lib/libhello_a-unistd.o -MD -MP -MF lib/.deps/libhello_a-unistd.Tpo -c -o lib/libhello_
a-unistd.o `test -f 'lib/unistd.c' || echo './'`lib/unistd.c
mv -f lib/.deps/libhello_a-unistd.Tpo lib/.deps/libhello_a-unistd.Po
gcc -DLOCALEDIR=\"/usr/local/share/locale\" -DHAVE_CONFIG_H -I. -Ilib -I./lib -Isrc -I./src -
Wno-cast-qual -Wno-conversion -Wno-float-equal -Wno-sign-compare -Wno-undef -Wno-unused-function
-Wno-unused-parameter -Wno-float-conversion -Wimplicit-fallthrough -Wno-pedantic -Wno-sign-conv
ersion -Wno-type-limits -Wno-unused-const-variable -Wno-unsuffixed-float-constants -Wno-error -g
-O2 -MT lib/unistr/libhello_a-u8-mbtoucr.o -MD -MP -MF lib/unistr/.deps/libhello_a-u8-mbtoucr.T
po -c -o lib/unistr/libhello_a-u8-mbtoucr.o `test -f 'lib/unistr/u8-mbtoucr.c' || echo './'`lib/
unistr/u8-mbtoucr.c
mv -f lib/unistr/.deps/libhello_a-u8-mbtoucr.Tpo lib/unistr/.deps/libhello_a-u8-mbtoucr.Po
gcc -DLOCALEDIR=\"/usr/local/share/locale\" -DHAVE_CONFIG_H -I. -Ilib -I./lib -Isrc -I./src -
Wno-cast-qual -Wno-conversion -Wno-float-equal -Wno-sign-compare -Wno-undef -Wno-unused-function
-Wno-unused-parameter -Wno-float-conversion -Wimplicit-fallthrough -Wno-pedantic -Wno-sign-conv
ersion -Wno-type-limits -Wno-unused-const-variable -Wno-unsuffixed-float-constants -Wno-error -g
-O2 -MT lib/unistr/libhello_a-u8-uctomb.o -MD -MP -MF lib/unistr/.deps/libhello_a-u8-uctomb.Tpo
-c -o lib/unistr/libhello_a-u8-uctomb.o `test -f 'lib/unistr/u8-uctomb.c' || echo './'`lib/unis
tr/u8-uctomb.c
mv -f lib/unistr/.deps/libhello_a-u8-uctomb.Tpo lib/unistr/.deps/libhello_a-u8-uctomb.Po
gcc -DLOCALEDIR=\"/usr/local/share/locale\" -DHAVE_CONFIG_H -I. -Ilib -I./lib -Isrc -I./src -
Wno-cast-qual -Wno-conversion -Wno-float-equal -Wno-sign-compare -Wno-undef -Wno-unused-function
-Wno-unused-parameter -Wno-float-conversion -Wimplicit-fallthrough -Wno-pedantic -Wno-sign-conv
ersion -Wno-type-limits -Wno-unused-const-variable -Wno-unsuffixed-float-constants -Wno-error -g
-O2 -MT lib/unistr/libhello_a-u8-uctomb-aux.o -MD -MP -MF lib/unistr/.deps/libhello_a-u8-uctomb
-aux.Tpo -c -o lib/unistr/libhello_a-u8-uctomb-aux.o `test -f 'lib/unistr/u8-uctomb-aux.c' || ec
ho './'`lib/unistr/u8-uctomb-aux.c
```

```
rm -f lib/libhello.a
ar cr lib/libhello.a lib/libhello_a-basename-lgpl.o lib/libhello_a-c-ctype.o lib/libhello_a-c-st
rcasecmp.o lib/libhello_a-c32isalnum.o lib/libhello_a-c32isalpha.o lib/libhello_a-c32isblank.o l
ib/libhello_a-c32iscntrl.o lib/libhello_a-c32isdigit.o lib/libhello_a-c32isgraph.o lib/libhello_
a-c32islower.o lib/libhello_a-c32isprint.o lib/libhello_a-c32ispunct.o lib/libhello_a-c32isspace
.o lib/libhello_a-c32isupper.o lib/libhello_a-c32isxdigit.o lib/libhello_a-c32tolower.o lib/libh
ello_a-c32width.o lib/libhello_a-cloexec.o lib/libhello_a-close-stream.o lib/libhello_a-closeou
t.o lib/libhello_a-dirname.o lib/libhello_a-basename.o lib/libhello_a-dirname-lgpl.o lib/libhell
o_a-stripslash.o lib/libhello_a-exitfail.o lib/libhello_a-fcntl.o lib/libhello_a-fd-hook.o lib/
libhello_a-fseterr.o lib/libhello_a-getprogname.o lib/libhello_a-hard-locale.o lib/libhell
o_a-ialloc.o lib/libhello_a-localcharset.o lib/libhello_a-malloca.o lib/libhello_a-mbchar
.o lib/libhello_a-mbrtoc32.o lib/libhello_a-mbrtowc.o lib/libhello_a-mbslen.o lib/libhello_a-mb
srtowcs.o lib/libhello_a-mbsstr.o lib/libhello_a-mbszero.o lib/libhello_a-mbuiter.o lib/libhell
o_a-mbuiterf.o lib/libhello_a-memeq.o lib/libhello_a-progname.o lib/libhello_a-propername.o
lib/libhello_a-quotearg.o lib/libhello_a-realloc.o lib/libhello_a-reallocarray.o lib/libhello_a-
setlocale_null.o lib/libhello_a-setlocale_null-unlocked.o lib/libhello_a-stat-time.o lib/li
bhello_a-stdlib.o lib/libhello_a-streq.o lib/libhello_a-striconv.o lib/libhello_a-strnlen1.o
lib/libhello_a-strnul.o lib/libhello_a-trim.o lib/unicase/libhello_a-tolower.o lib/unictype/libh
ello_a-ctype_alnum.o lib/unictype/libhello_a-ctype_alpha.o lib/unictype/libhello_a-ctype_blank.o
lib/unictype/libhello_a-ctype_cntrl.o lib/unictype/libhello_a-ctype_digit.o lib/unictype/libhel
lo_a-ctype_graph.o lib/unictype/libhello_a-ctype_lower.o lib/unictype/libhello_a-ctype_print.o l
ib/unictype/libhello_a-ctype_punct.o lib/unictype/libhello_a-ctype_space.o lib/unictype/libhell
o_a-ctype_upper.o lib/unictype/libhello_a-ctype_xdigit.o lib/libhello_a-unistd.o lib/unistr/libh
ello_a-u8-mbtoucr.o lib/unistr/libhello_a-u8-uctomb.o lib/unistr/libhello_a-u8-uctomb-aux.o lib/u
niwidth/libhello_a-width.o lib/libhello_a-version-etc.o lib/libhello_a-version-etc-fsf.o lib/lib
hello_a-wctype-h.o lib/libhello_a-xmalloc.o lib/libhello_a-xalloc-die.o lib/libhello_a-xstricon
v.o lib/libhello_a-mbsrtowcs-state.o
ranlib lib/libhello.a
gcc -g -O2 -WL,--as-needed -o hello src/hello.o lib/libhello.a
make[2]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[1]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
```

Potem można sprawdzić, czy kompilacja przebiegła pomyślnie, za pomocą komendy make check.

```

pawel@myubuntu:~/Pobrane/hello-2.12.3$ make check
if test -d ./.git
    && test -e ./.gitmodules
    && git --version >/dev/null 2>&1; then
    cd . &&
    git submodule --quiet foreach
        'test "$(git rev-parse "$sha1")"
            = "$(git merge-base origin "$sha1")"'
        || { echo 'maint.mk: found non-public submodule commit' >&2;
            exit 1; };
else
    : ;
fi
make check-recursive
make[1]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
Making check in po
make[2]: Entering directory '/home/pawel/Pobrane/hello-2.12.3/po'
make[2]: Nothing to be done for 'check'.
make[2]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3/po'
make[2]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
make check-TESTS
make[3]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
make[4]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
PASS: tests/atexit-1
PASS: tests/greeting-1
SKIP: tests/greeting-2
PASS: tests/hello-1
PASS: tests/last-1
PASS: tests/operand-1
PASS: tests/traditional-1
PASS: tests/operand-1
PASS: tests/traditional-1
=====
Testsuite summary for GNU Hello 2.12.3
=====
# TOTAL: 7
# PASS: 6
# SKIP: 1
# XFAIL: 0
# FAIL: 0
# XPASS: 0
# ERROR: 0
=====
make[4]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[3]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[2]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[1]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
pawel@myubuntu:~/Pobrane/hello-2.12.3$

```

Następnie należy zainstalować pakiet, za pomocą polecenia `sudo make install`.

```
pawel@myubuntu:~/Pobrane/hello-2.12.3$ sudo make install
[sudo] hasło użytkownika pawel:
make install-recursive
make[1]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
Making install in po
make[2]: Entering directory '/home/pawel/Pobrane/hello-2.12.3/po'
installing ast.gmo as /usr/local/share/locale/ast/LC_MESSAGES/hello.mo
installing bg.gmo as /usr/local/share/locale/bg/LC_MESSAGES/hello.mo
installing ca.gmo as /usr/local/share/locale/ca/LC_MESSAGES/hello.mo
installing cs.gmo as /usr/local/share/locale/cs/LC_MESSAGES/hello.mo
installing da.gmo as /usr/local/share/locale/da/LC_MESSAGES/hello.mo
installing de.gmo as /usr/local/share/locale/de/LC_MESSAGES/hello.mo
installing el.gmo as /usr/local/share/locale/el/LC_MESSAGES/hello.mo
installing eo.gmo as /usr/local/share/locale/eo/LC_MESSAGES/hello.mo
installing es.gmo as /usr/local/share/locale/es/LC_MESSAGES/hello.mo
installing et.gmo as /usr/local/share/locale/et/LC_MESSAGES/hello.mo
installing eu.gmo as /usr/local/share/locale/eu/LC_MESSAGES/hello.mo
installing fa.gmo as /usr/local/share/locale/fa/LC_MESSAGES/hello.mo
installing fi.gmo as /usr/local/share/locale/fi/LC_MESSAGES/hello.mo
installing fr.gmo as /usr/local/share/locale/fr/LC_MESSAGES/hello.mo
installing ga.gmo as /usr/local/share/locale/ga/LC_MESSAGES/hello.mo
installing gl.gmo as /usr/local/share/locale/gl/LC_MESSAGES/hello.mo
installing he.gmo as /usr/local/share/locale/he/LC_MESSAGES/hello.mo
installing hr.gmo as /usr/local/share/locale/hr/LC_MESSAGES/hello.mo
installing hu.gmo as /usr/local/share/locale/hu/LC_MESSAGES/hello.mo
installing id.gmo as /usr/local/share/locale/id/LC_MESSAGES/hello.mo
installing it.gmo as /usr/local/share/locale/it/LC_MESSAGES/hello.mo
installing ja.gmo as /usr/local/share/locale/ja/LC_MESSAGES/hello.mo
installing ka.gmo as /usr/local/share/locale/ka/LC_MESSAGES/hello.mo
installing ko.gmo as /usr/local/share/locale/ko/LC_MESSAGES/hello.mo
installing lv.gmo as /usr/local/share/locale/lv/LC_MESSAGES/hello.mo
```



```

installing tr.gmo as /usr/local/share/locale/tr/LC_MESSAGES/hello.mo
installing uk.gmo as /usr/local/share/locale/uk/LC_MESSAGES/hello.mo
installing vi.gmo as /usr/local/share/locale/vi/LC_MESSAGES/hello.mo
installing zh_CN.gmo as /usr/local/share/locale/zh_CN/LC_MESSAGES/hello.mo
installing zh_TW.gmo as /usr/local/share/locale/zh_TW/LC_MESSAGES/hello.mo
if test "hello" = "gettext-tools"; then \
  /usr/bin/mkdir -p /usr/local/share/gettext/po; \
  for file in Makefile.in.in remove-potcdate.sin quot.sed boldquot.sed en@quot.header en@boldquo
t.header insert-header.sin Rules-quot Makevars.template; do \
    /usr/bin/install -c -m 644 ./\$file \
      /usr/local/share/gettext/po/\$file; \
  done; \
  for file in Makevars; do \
    rm -f /usr/local/share/gettext/po/\$file; \
  done; \
else \
  : ; \
fi
make[2]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3/po'
make[2]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
make[3]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
/usr/bin/mkdir -p '/usr/local/bin'
/usr/bin/install -c hello '/usr/local/bin'
/usr/bin/mkdir -p '/usr/local/share/info'
/usr/bin/install -c -m 644 ./doc/hello.info '/usr/local/share/info'
install-info --info-dir='/usr/local/share/info' '/usr/local/share/info/hello.info'
/usr/bin/mkdir -p '/usr/local/share/man/man1'
/usr/bin/install -c -m 644 hello.1 '/usr/local/share/man/man1'
make[3]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[2]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
make[1]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
pawel@myubuntu:~/Pobrane/hello-2.12.3$

```

Potem można sprawdzić, czy instalacja przebiegła pomyślnie, za pomocą komendy `sudo make installcheck`.

```

pawel@myubuntu:~/Pobrane/hello-2.12.3$ sudo make installcheck
Making installcheck in po
make[1]: Entering directory '/home/pawel/Pobrane/hello-2.12.3/po'
make[1]: Nothing to be done for 'installcheck'.
make[1]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3/po'
make[1]: Entering directory '/home/pawel/Pobrane/hello-2.12.3'
make[1]: Nothing to be done for 'installcheck-am'.
make[1]: Leaving directory '/home/pawel/Pobrane/hello-2.12.3'
pawel@myubuntu:~/Pobrane/hello-2.12.3$

```

Po całym powyższym procesie można już uruchomić program.

```

pawel@myubuntu:~/Pobrane/hello-2.12.3$ hello
Witaj, świecie!
pawel@myubuntu:~/Pobrane/hello-2.12.3$

```

Wnioski:

Instalacja przebiegła sprawnie i pomyślnie. Taki proces działania to przyjemność pracy.